

- 1- A flat plate with an angle of attack of 10 degrees is at a supersonic flow at $M=3.5$. From linear theory and shock-expansion theory, find the C_l and C_d of this airfoil and compare both data together.
- 2- A triangle with both angles of 7.5 degrees is located at $M=3.5$ in a supersonic flow at $\alpha=10$ degrees. From linear theory and shock-expansion theory, find the C_l and C_d of this airfoil and compare both data together.
- 3- A diamond airfoil with all angles of 7.5 degrees is located at $M=3.5$ in a supersonic flow at $\alpha=10$ degrees. From linear theory and shock-expansion theory, find the C_l and C_d of this airfoil and compare both data together.
- 4- Compare the above three airfoils together and discuss the reasons for the higher lift and lower drag of the airfoil that is the best case.